

LAGUNA STATE POLYTECHNIC UNIVERSITY

ITEP 414 - System Administration and Maintenance

ENTERPRISE INFRASTRUCTURE PLAN

ABC Startup Solutions

Week 2 Portfolio Project

Prepared by: Cedrick Gayoso

Program: Bachelor of Science in Information Technology

Course: ITEP 414 - System Administration and Maintenance

Week: 2

PROJECT OVERVIEW

Every successful IT infrastructure begins with proper planning. Before purchasing computers, installing servers, configuring networks, or deploying cloud services, a System Administrator must first understand the organization's business requirements and design an infrastructure that supports business operations.

In this project, I assume the role of a Junior System Administrator assigned to prepare the initial IT Infrastructure Plan of a newly established startup company. The final output is presented as a professional document that could be submitted to an IT Manager or Company Executive.

PROJECT SCENARIO

I have recently been hired as the Junior System Administrator of ABC Startup Solutions, a newly established software development company. The company has 20 employees and occupies a single office floor. Management has requested a complete IT Infrastructure Plan before purchasing equipment.

Department	Employees
Information Technology	5
Human Resources	4
Finance	5
Sales	6
TOTAL	20

The company currently has:

- No computers
- No server
- No network
- No internet infrastructure
- No security policies

The task is to design the startup company's initial IT infrastructure from scratch.

PART 1 - COMPANY PROFILE

Item	Details
Company Name	ABC Startup Solutions
Nature of Business	Newly established software development company.
Company Vision	To become a trusted software development company that delivers reliable and useful solutions while maintaining a secure and professional IT environment.

Item	Details
Company Mission	To develop practical software solutions, support business operations through dependable technology, and protect company information through organized and responsible IT practices.
Office Location (Fictional)	Calamba City, Laguna, Philippines (fictional office location).

Organizational Structure

ABC Startup Solutions - Organizational Structure

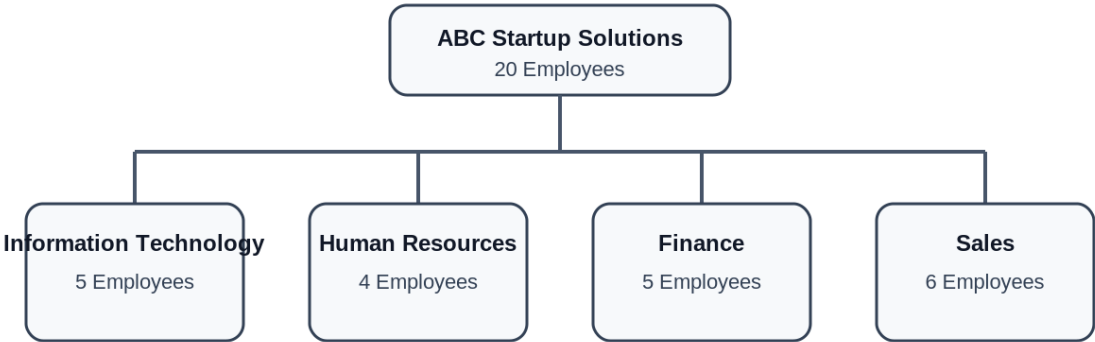


Figure 1. Organizational structure of ABC Startup Solutions.

Employee Distribution

Department	Employees
Information Technology	5
Human Resources	4
Finance	5
Sales	6
TOTAL	20

PART 2 - ENTERPRISE HARDWARE INVENTORY

Asset ID	Hardware	Quantity	Department	Purpose / Justification
HW-001	Desktop Computers	9	HR (4), Finance (5)	Provides fixed workstations for departments that mainly work inside the office.
HW-002	Laptops	11	IT (5), Sales (6)	Provides mobility for development, technical work, presentations, and sales activities.

Asset ID	Hardware	Quantity	Department	Purpose / Justification
HW-003	Server	1	IT	Provides centralized services, shared resources, and internal system support.
HW-004	Router	1	IT / Network	Connects the company network to the internet connection and directs network traffic.
HW-005	Switch	1	IT / Network	Connects wired computers, the server, printer, and other network devices on the office floor.
HW-006	Printer	2	Shared	Allows departments to print and scan documents without requiring one printer per department.
HW-007	UPS	2	IT / Network	Provides short-term backup power for the server and important network equipment.
HW-008	Wireless Access Point	2	Shared Office	Provides wireless access across the single office floor and improves coverage compared with one access point.
HW-009	NAS Storage	1	IT	Provides centralized storage for shared files and local backup copies.
HW-010	External Backup Drive	2	IT	Provides additional backup copies that can be disconnected and stored safely.
HW-011	Monitors	20	All Departments	Provides one external display for each employee workstation.

Quantity Justification: The company has 20 employees. HR and Finance are assigned 9 desktops, while IT and Sales are assigned 11 laptops. Shared infrastructure such as the server, router, switch, NAS, access points, printers, UPS units, and backup drives is sized for one office floor and the current 20-person organization.

PART 3 - ENTERPRISE SOFTWARE INVENTORY

Software	Version	License	Purpose
Windows 11 Pro	Current supported release	Commercial	Primary operating system for employee computers and business workstation management.
Ubuntu Server	Current LTS release	Open-source	Server operating system for centralized services and administration.
Microsoft Office	Current supported release	Commercial	Creates and manages documents, spreadsheets, presentations, and office files.
VS Code	Current stable release	Free	Code editor for software development and technical files.
Git	Current stable release	Open-source	Tracks changes in source code and technical documentation.
GitHub Desktop	Current stable release	Free	Provides a graphical interface for Git and GitHub repository work.
VirtualBox	Current stable release	Free / Open-source	Creates virtual machines for testing operating systems and configurations.
Google Chrome	Current stable release	Free	Web browser for cloud services, research, web applications, and business access.

Software	Version	License	Purpose
Microsoft Defender	Built into Windows 11	Included with Windows	Provides antivirus and malware protection for Windows computers.
AnyDesk	Current stable release	Business license	Supports authorized remote troubleshooting and technical assistance.
7-Zip	Current stable release	Free / Open-source	Compresses and extracts files for storage, transfer, and technical work.

PART 4 - ENTERPRISE NETWORK INVENTORY

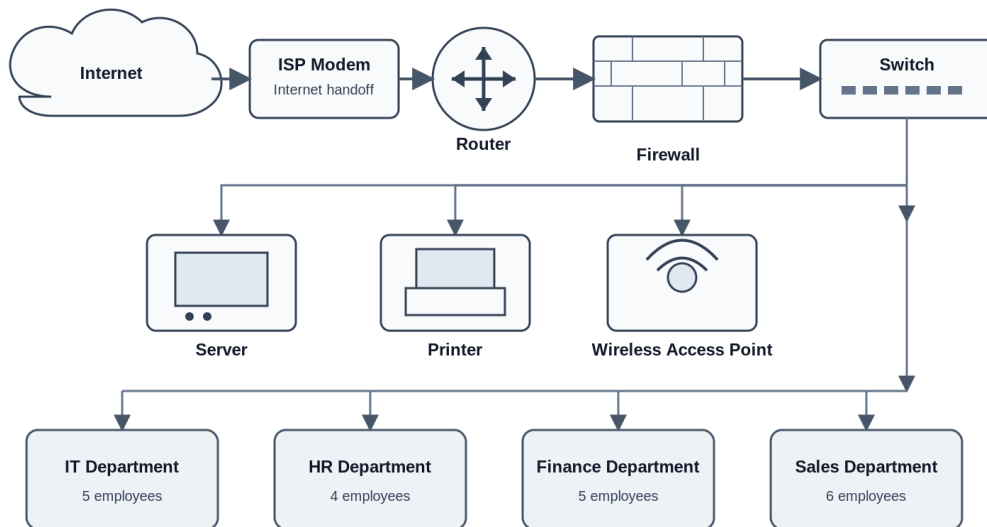
Asset ID	Network Equipment	Quantity	Location	Purpose / Justification
NET-001	ISP Modem	1	Provider handoff	Connects the office to the internet service.
NET-002	Router	1	Network area	Routes traffic between the internet connection and the company network.
NET-003	Firewall	1	Network area	Filters traffic and protects the internal network from unauthorized access.
NET-004	Switch	1	Network area	Connects wired computers, the server, printer, and access points.
NET-005	Access Point	2	Office ceiling / shared area	Provides wireless network coverage across the single office floor.
NET-006	Patch Panel	1	Network area	Organizes and terminates office network cable connections.
NET-007	CAT6 Cables	1 lot	Office cabling	Connects employee work areas and network devices through wired Ethernet.
NET-008	RJ45 Connectors	1 lot	IT stock	Provides connectors for CAT6 network cable termination and maintenance.

PART 5 - ENTERPRISE NETWORK DIAGRAM

The network topology below includes the Internet, ISP Modem, Router, Firewall, Switch, Wireless Access Point, Server, Printer, and the Information Technology, Human Resources, Finance, and Sales departments. The connections show the logical path from the internet connection to the internal network and office devices.

ABC Startup Solutions - Enterprise Network Topology

Student: Cedrick Gayoso



Logical flow: Internet -> ISP Modem -> Router -> Firewall -> Switch -> Network devices and departments

Figure 2. ABC Startup Solutions Enterprise Network Topology

PART 6 - SYSTEM ADMINISTRATION ROLES

Helpdesk Technician

Area	Details
Responsibilities	Receives user concerns, troubleshoots common computer and software problems, assists with device and account setup, documents issues, and escalates problems when needed.
Skills	Communication, customer support, troubleshooting, basic hardware and software knowledge, and basic networking knowledge.
Common Tools	Ticketing system, remote support tool, operating system utilities, command-line tools, and documentation.
Certifications	CompTIA A+ and other entry-level technical support certifications.

Network Administrator

Area	Details
Responsibilities	Maintains the router, firewall, switch, wireless access points, and network connections; monitors availability; applies network settings; and troubleshoots connectivity problems.
Skills	TCP/IP, routing, switching, wireless networking, network security, structured troubleshooting, and documentation.
Common Tools	Network monitoring tools, command-line utilities, packet analysis tools, device management interfaces, and network diagrams.

Area	Details
Certifications	CCNA and CompTIA Network+ are examples of networking certifications.

Linux System Administrator

Area	Details
Responsibilities	Installs and manages the Ubuntu Server, maintains users and permissions, applies updates, checks services and logs, manages storage, and supports backups.
Skills	Linux command line, user and permission management, service administration, basic scripting, storage, logging, and troubleshooting.
Common Tools	Terminal, SSH, system service tools, log tools, text editors, scripting tools, and backup utilities.
Certifications	CompTIA Linux+ and Red Hat Certified System Administrator are examples of Linux administration certifications.

Cloud Administrator

Area	Details
Responsibilities	Manages approved cloud services and resources, controls access, monitors cloud systems, supports cloud storage and backups, and helps maintain cloud security.
Skills	Cloud concepts, identity and access management, virtual networking, monitoring, security, backup, and basic automation.
Common Tools	Cloud management console, command-line interface, monitoring tools, access-management tools, and backup tools.
Certifications	Entry-level or administrator certifications from major cloud platforms are examples of relevant certifications.

How the Four Professionals Work Together

The four professionals work together by handling different parts of the same IT environment. The Helpdesk Technician receives user concerns and resolves common problems first. Network-related issues are handled by the Network Administrator, server-related issues are handled by the Linux System Administrator, and cloud-related issues are handled by the Cloud Administrator. They share information, documentation, and troubleshooting results so that problems can be resolved efficiently and the organization can keep its systems available, organized, and secure.

PART 7 - INFRASTRUCTURE RECOMMENDATIONS

Internet Provider

Use a reliable business fiber internet provider available at the fictional office location. A business connection is appropriate because 20 employees will depend on web access, software services, communication, downloads, and online work. A backup internet connection can be added when the company needs better availability.

Server Specifications

Use one business server with an 8-core processor, at least 32 GB of memory, mirrored solid-state storage for the operating system, additional storage for shared data, and at least two network interfaces. This provides enough capacity for centralized services while leaving room for future growth.

Backup Strategy

Use the NAS for regular local backups and keep additional copies on the two external backup drives. At least one external drive should be disconnected after backup so a separate copy remains available. Backup files should be checked and restored during testing to confirm that recovery works.

Security Recommendations

Place the firewall between the internet connection and the internal network, keep systems updated, limit user access to what is needed, protect administrator accounts, separate guest wireless access from company use when guest access is provided, and keep written records of important system changes and incidents.

Antivirus

Keep Microsoft Defender enabled and updated on all Windows 11 Pro computers. Regular scanning and security updates help protect employee devices from malware.

Password Policy

Require strong, unique passwords or passphrases for business accounts. Passwords should not be shared, reused across business accounts, or stored in unsecured files. Multi-factor authentication should be used for important accounts when available.

Expansion Plan

Keep spare switch ports, storage capacity, and room for additional devices. As the number of employees increases, add computers, software licenses, network ports, access points, storage, and server capacity based on the new requirements. Update the inventories and network diagram whenever major changes are made.

PART 8 - PERSONAL REFLECTION

Completing this Week 2 activity helped me understand that system administration begins with planning before any computer, server, or network device is purchased. I learned that the needs of the organization should guide technical decisions. The company has only 20 employees, but each department still needs the correct type of computer, software, network access, shared resources, security, and backup support. Preparing the hardware, software, and

network inventories also showed me that an inventory is more useful when every item has a clear quantity, department, purpose, and justification.

The most challenging part for me was the enterprise network diagram. I had to think about the correct order of the Internet, ISP Modem, Router, Firewall, Switch, Wireless Access Point, Server, Printer, and department connections. The diagram needed to be easy to understand while still showing a logical network flow. This task made me pay more attention to labels, organization, and the relationship between physical devices and the people who will use them.

Planning is important before deployment because it helps prevent missing equipment, unnecessary purchases, weak security, and poor organization. A complete plan allows the System Administrator to check whether the company has enough computers for all employees, enough network connections for office devices, a server for centralized services, storage for files, and backup equipment for recovery. Planning also makes security part of the design instead of something that is added only after a problem occurs.

This project will help me become a better System Administrator because it gave me practice in analyzing a business scenario and turning it into a complete infrastructure plan. I practiced preparing inventories, designing a network topology, explaining system administration roles, recommending backup and security practices, and documenting everything in a professional format. I also learned that system administration requires communication and teamwork because Helpdesk, Network, Linux, and Cloud professionals must share information when solving problems. In future activities, I will try to understand the requirements first, organize the plan clearly, consider security and recovery, and document important decisions before moving to implementation.

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